Final Presentation

Internship Program
Department of Biomedical Engineering (BME)
National Cheng Kung University (NCKU)

Presenter：Faiz Henri Kurniawan

Advisor： Prof. Che-Wei Lin

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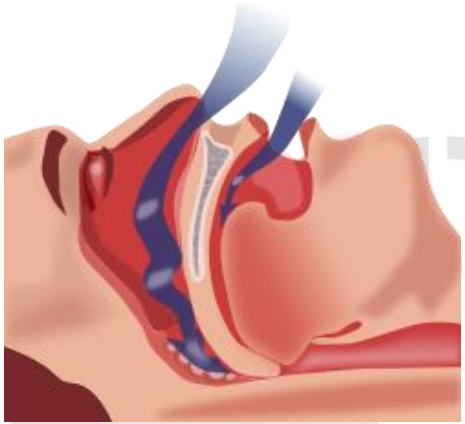
Outline

- Introduction
  - What is OSA
  - Problem of Diagnosis OSA
  - Research Gap
- Methodology
  - System Architecture
  - Feature Generation
  - Model's Architecture
  - Evaluation Parameters
- Result
  - Discussion and Conclusion
  - Future Work
- Webpage Work

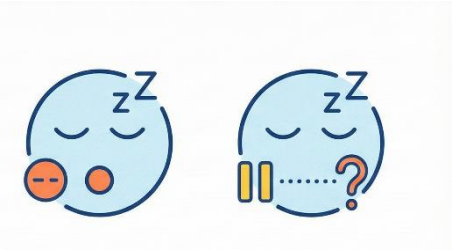
Obstructive Sleep Apnea (OSA)



- Obstructive Sleep Apnea (OSA) is a **sleep disorder characterized by repeated pauses or reductions in breathing during sleep**, which occur due to temporary collapse or obstruction of the upper airway, resulting in disrupted airflow despite ongoing breathing effort.

- Common symptoms

Loud or chronic snoring

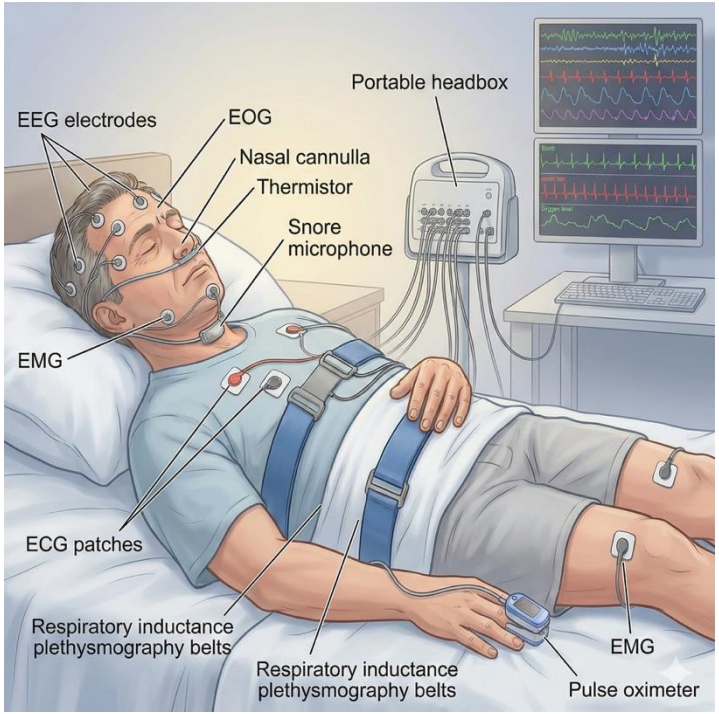


Observed breathing pauses during sleep

Excessive daytime sleepiness

Morning headache and persistent fatigue

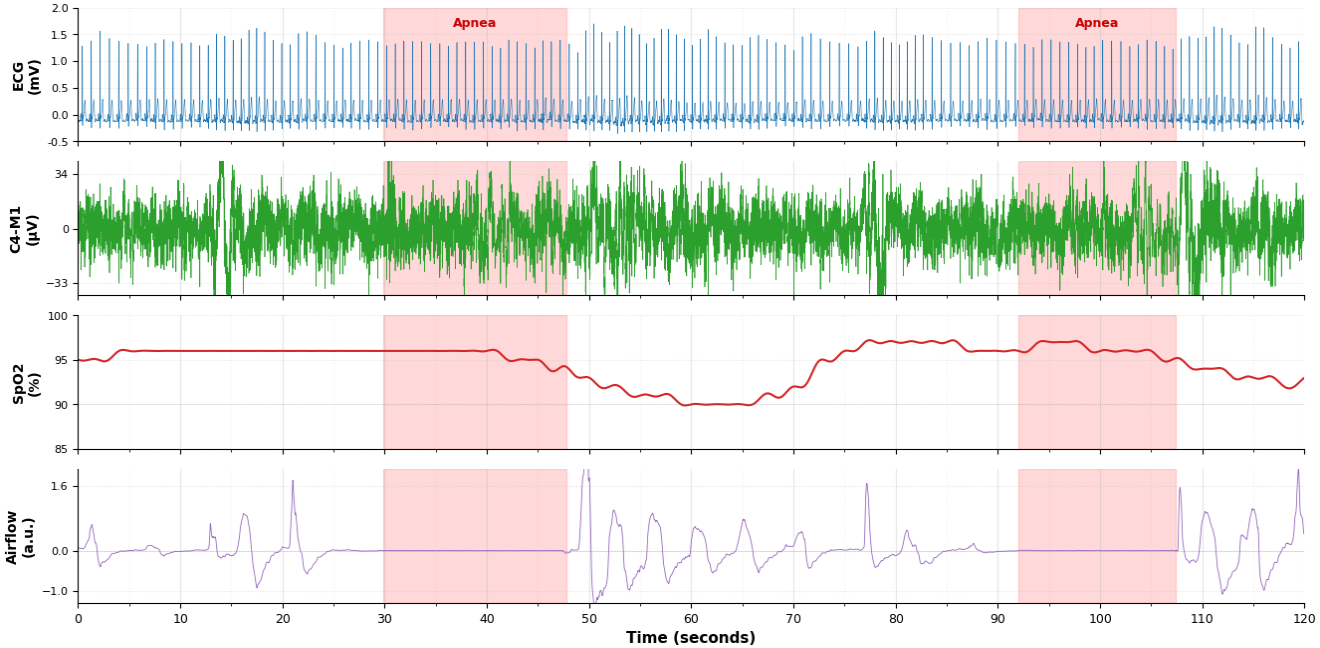
Diagnosing Obstructive Sleep Apnea



- **Polysomnography (PSG) is the clinical gold standard for diagnosing Obstructive Sleep Apnea because** it records multiple physiological signals during sleep, such as brain activity, oxygen levels, and muscle movement.
- However, PSG is expensive, complex, and requires patients to stay overnight in a sleep laboratory, which **limits its practicality for large-scale or long-term monitoring.**

Yifei Wang, Qi Liu, Fuli Min, Honghao Wang, "PSG-MAE: Robust Multitask Sleep Event Monitoring Using Multichannel PSG Reconstruction and Inter-Channel Contrastive Learning", *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol.34, pp.274-286, 2026.

Diagnosing Obstructive Sleep Apnea



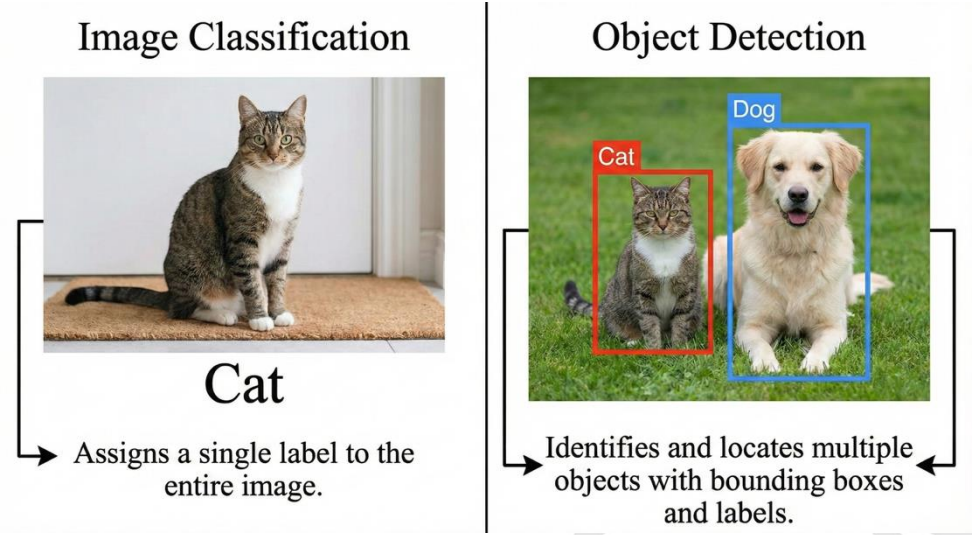
- ECG signals reflect cardiac activity and are modulated by autonomic nervous system regulation and respiration, which change during obstructive sleep apnea events.

Penzel T, Kantelhardt JW, Bartsch RP, Riedl M, Kraemer JF, Wessel N, Garcia C, Glos M, Fietze I and Schöbel C (2016) Modulations of Heart Rate, ECG, and Cardio-Respiratory Coupling Observed in Polysomnography. Front. Physiol. 7:460. doi: 10.3389/fphys.2016.00460

Research Gap: Why Object Detection?

Paper Title	Year	Dataset	Signal Type(s)	Window Size (sec)	Based Model	Evaluation method
Manish Sharma et al.	2018	Physionet	ECG	**60 sec**	LS-SVM	**Classification**
Kaiyin Zhu et al.	2019	Custom data	thoracic region	**30 sec**	Random Forest	**Classification**
Mads Olsen et al.	2019	SHHS, MESA	ECG	**5 min**	LSTM-RNN	**Classification**
Hemant Sharma and KK Sharma	2020	Physionet	ECG	**60 sec**	KNN, LDA	**Classification**
Hengji Qin et al.	2021	Physionet	ECG (RR Interval)	**60 sec**	1D-CNN, BiGRU	**Classification**

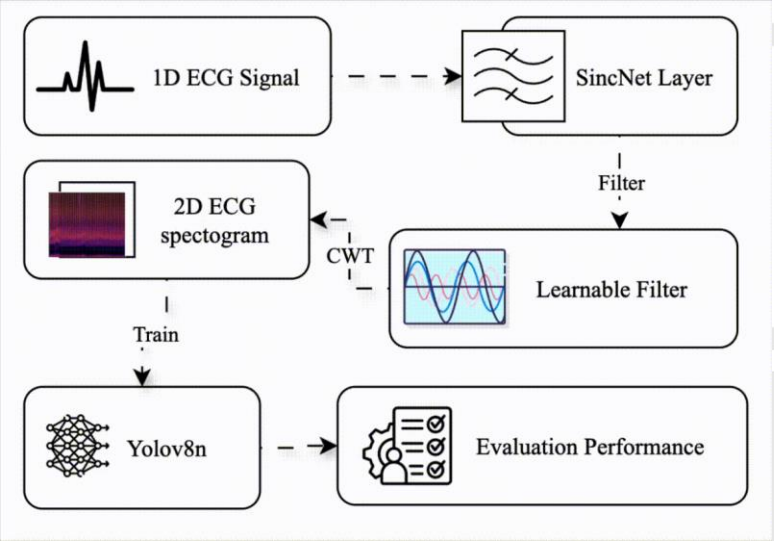
Note: SHHS: The Sleep Heart Health Database ; MESA : Multi-Ethnic Study of Atherosclerosis ; Physionet : Physionet's Apnea-ECG database



While most ECG-based OSA studies rely on epoch-level classification that identifies what occurs within a fixed segment, object detection enables apnea events to be treated as temporal objects, allowing both their occurrence and precise timing to be identified in continuous ECG signals.

Motivation

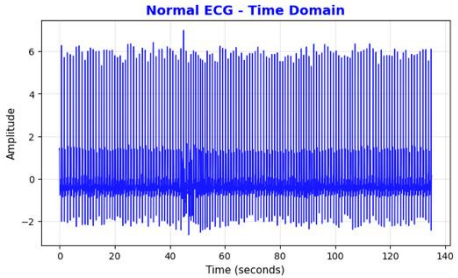
- Previous studies have demonstrated that **obstructive sleep apnea events can be detected as temporal objects using object detection** on ECG spectrograms, where ECG signals are transformed into 2D time–frequency representations [1].
- This study adapts the established **object-detection framework and explores frequency** representation by learning frequency bands from 1D ECG signals using SincNet.[3]
- Detection results using SincNet-derived frequency bands are compared with those using predefined frequency ranges from prior ECG-based studies[2].

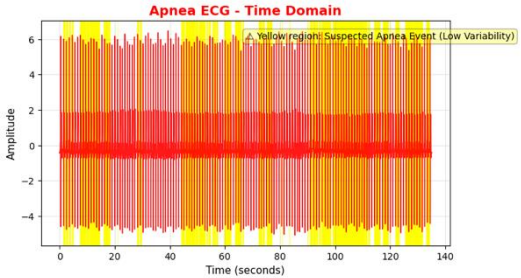


[1] P. Nipattanon & C. Lin, "Event-Based detection of obstructive sleep apnea using YOLOV8n," in *World Congress on Medical Physics and Biomedical Engineering*, 2025.
[2] C. Lin, Y. Wang, et al., "Sleep Apnea Classification Using Machine-Learning and Bag-of-Features," *Journal of Clinical Medicine*, vol. 11, no. 1, 2021.
[3] Interpretability and optimisation of CNN based on SINC-Convolution, *IEEE Journals & Magazine*, 2023.

Dataset

- Dataset: 50 subjects from NCKUHSC
  - 50 subjects with 27,000-sample ECG segments at 200 Hz sampling rate
  - Source: Sleep apnea dataset with labeled ECG recordings from 50 subjects
- ECG Duration:
  - 135 seconds per segment (27,000 samples at 200 Hz)
  - Class Distribution: Normal (Class 0) and Apnea (Class 1)
- Data Preprocessing:
  - Segmentation into 27,000-sample windows from raw recordings
  - Storage Format: NumPy binary format (one file per segment)
  - Total Segments: Approximately 3,000 segments across all subjects
- Data Split Strategy:
  - 64% training, 14% validation, 16% test





segmented_npy > s50_1460_label1.npy

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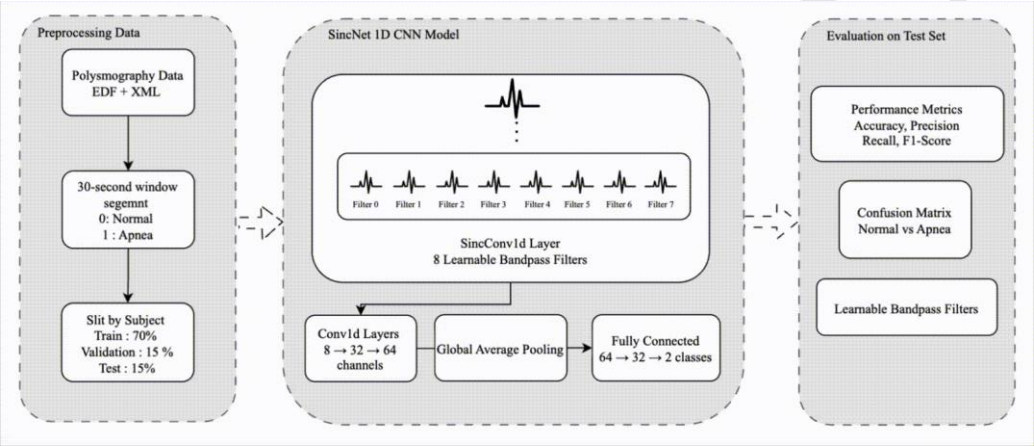
```


Note: **NCKUHSC: National Cheng Kung University Hospital Sleep Center.**

SincNet Layer

- SincNet is a 1D CNN that uses learnable band-pass filters, enabling interpretable frequency learning from raw signals.
- Filter cutoff frequencies are learned directly, allowing SincNet to capture dominant frequency characteristics in ECG signals.
- Prior studies show that sinc-based filters can represent physiologically meaningful ECG frequency bands.
- In this study, **<u>SincNet is used on 1D ECG segments to explore frequency bands</u>**, which are then compared with predefined frequency ranges.



Śmigiel, S.; Pałczyński, K.; Ledziński, D. ECG Signal Classification Using Deep Learning Techniques Based on the PTB-XL Dataset. *Entropy* **2021**, *23*, 1121. https://doi.org/10.3390/e23091121

Training Configuration

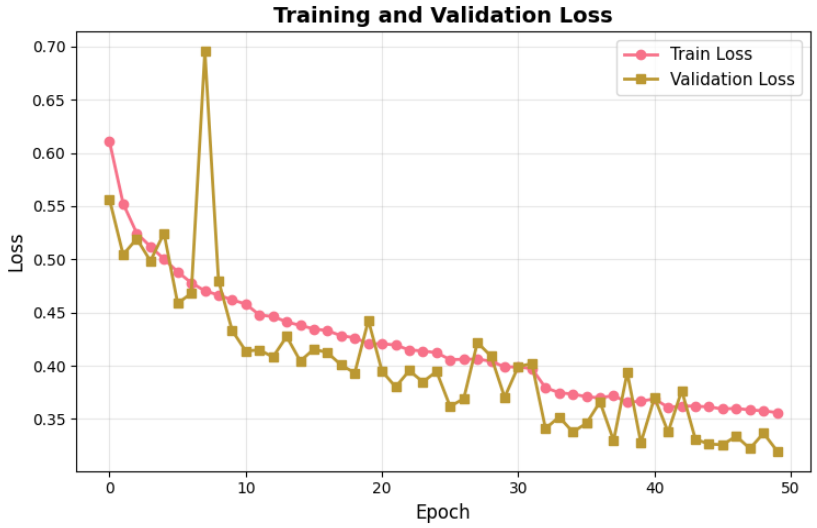
Data Split

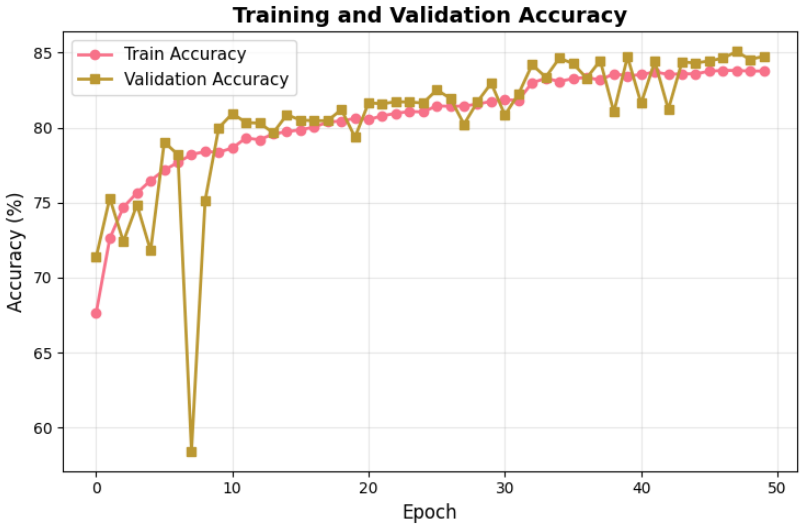
Split	Subjects	Subject IDs	Samples	% of Total Samples
Training	35	1, 2, 4, 5, 6, 7, 9, 10, 12, 13, 14, 16, 17, 18, 20, 22, 25, 26, 27, 28, 30, 31, 32, 33, 34, 35, 37, 38, 40, 42, 45, 46, 47, 48, 49	56,908	70.0%
Validation	7	3, 11, 23, 24, 36, 41, 44	11,015	14%
Test	8	8, 15, 19, 21, 29, 39, 43, 50	12,570	16%
Total	**50**	–	**80,493**	**100%**

Training Configuration

Parameter	Value	Details
Epochs	50	Number of training iterations
Learning Rate	0.001	Step size for weight updates
Optimizer	Adam	Adaptive optimization algorithm
Loss Function	Cross Entropy Loss	Measures classification error
Learning Rate Scheduler	ReduceLROnPlateau	Reduces LR when validation plateaus

Training and Validation Plott





- Training loss was finalized at 0.3560 with a training accuracy of 83.78 percent. Validation loss reached 0.3199 with a validation accuracy of 84.76 percent, and the best accuracy of 85.07 percent was achieved at epoch forty-eight.

Model

Layer	Type	Output Channels	Kernel / Params	Notes
sinc	SincConv1d	8 filters	Kernel size 252	Learnable band-pass filters
pool1	MaxPool1d	–	Kernel 4, stride 4	Downsampling
conv2	Conv1d	32	Kernel 5, padding 2	Extract mid-level features
bn2	BatchNorm1d	–	–	Stabilizes training
pool2	MaxPool1d	–	Kernel 4, stride 4	Downsampling
conv3	Conv1d	64	Kernel 5, padding 2	Higher-level representation
bn3	BatchNorm1d	–	–	Normalize activations
pool3	MaxPool1d	–	Kernel 4, stride 4	Downsampling
gap	AdaptiveAvgPool1d	1	–	Global feature aggregation
fc1	Linear	32	64 → 32	Fully connected head
dropout	Dropout	–	p = 0.5	Regularization
fc2	Linear	2	32 → 2	Output logits (Normal / Apnea)

This model uses a SincConv front-end with eight learnable band-pass filters, followed by two convolutional blocks and global pooling. It is lightweight, with only thirteen thousand nine hundred seventy parameters.

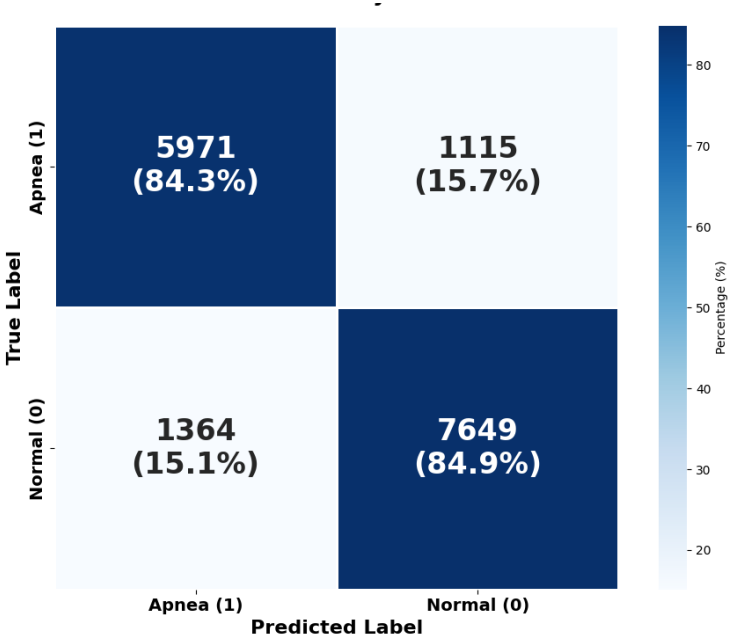
Evaluation Matrix

Classification (Per Class)

Class	Precision	Recall	F1-Score	Accuracy
Class 0 (Normal)	0.8728	0.8487	0.8606	84.87%
Class 1 (Apnea)	0.8140	0.8426	0.8281	84.26%

Overall Test Metrics

Metric	Value
Accuracy	**84.60%**
Precision	**0.8469**
Recall	**0.8460**
F1-Score	**0.8463**



Learned Frequency Bands

- Before vs After Training

Filter	Initial Low (Hz)	Trained Low (Hz)	Initial High (Hz)	Trained High (Hz)
Filter 0	1.00	0.50	6.00	6.52
Filter 1	5.21	5.15	10.21	10.26
Filter 2	9.43	9.31	14.43	14.22
Filter 3	13.64	13.56	18.64	18.21
Filter 4	17.86	17.49	22.86	22.74
Filter 5	22.07	22.05	27.07	27.30
Filter 6	26.29	26.11	31.29	31.37
Filter 7	30.50	31.24	35.50	36.30

Adapted System Architecture

Predefined Frequency Range	SincNet learnable Frequency Range
- 0.8 - 10	- 0.50 – 6.52
- 0.1- 50	- 5.15 - 10.26
- 8 - 50	- 9.31 - 14.22
- 0.001- 0.08	- 13.56 - 18.21

Feature Generation

90 Second segment ECG signal → Continues Wavelet Transfrom: Morlet Wavelet → 2D ECG spectogram

Evaluation Performance

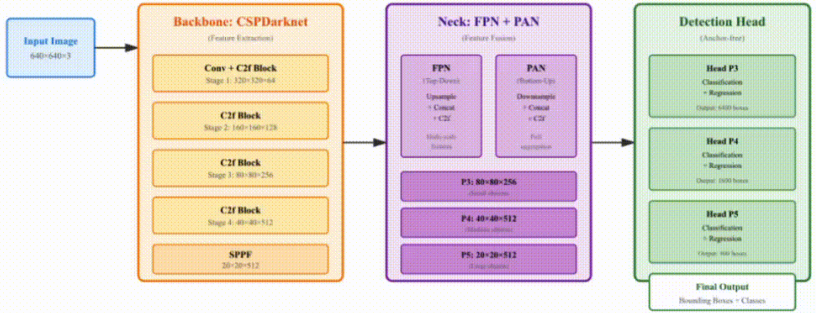
5 Fold Cross-Validation

Training	Training	Training	Training	Testing
Training	Training	Training	Testing	Training
Training	Training	Testing	Training	Training
Training	Testing	Training	Training	Training
Testing	Training	Training	Training	Training

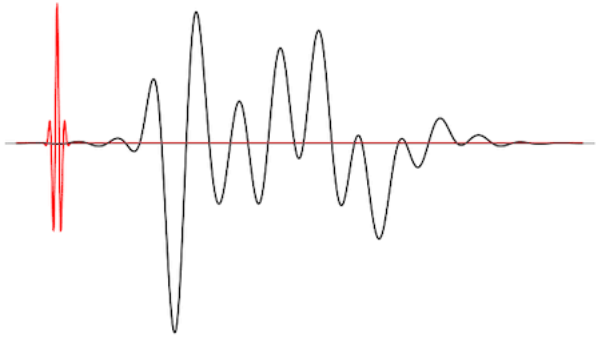
- Subject-Wise 5-Fold Cross-Validation
- Leave One Subject Out Cross - Validation

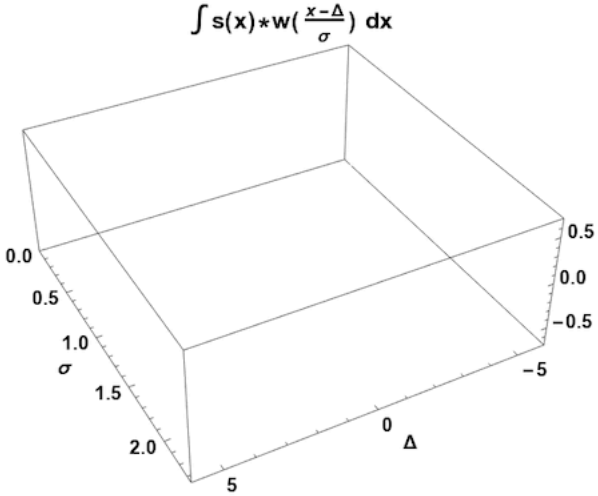
Training model : YOLOv8 nano

YOLOv8n Architecture



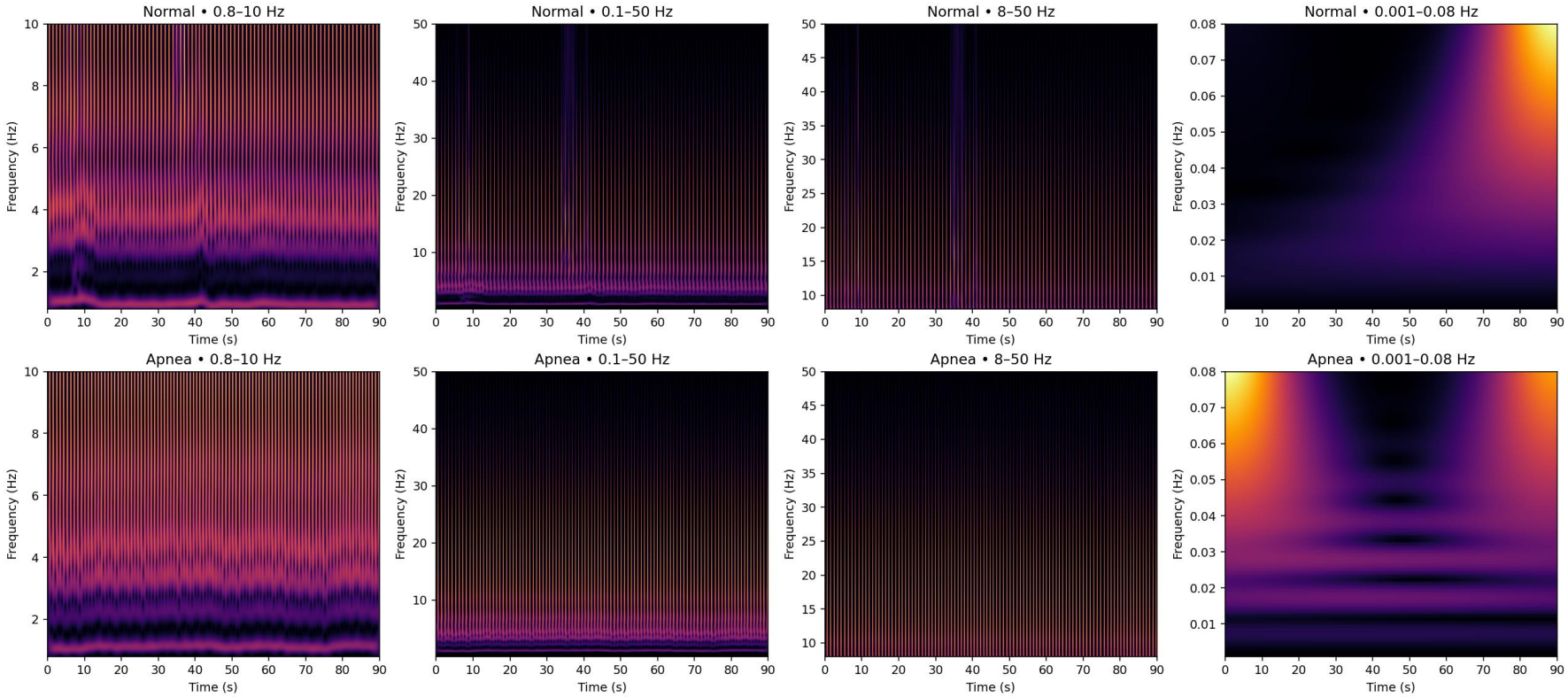
Continuous Wavelet Transform (CWT)



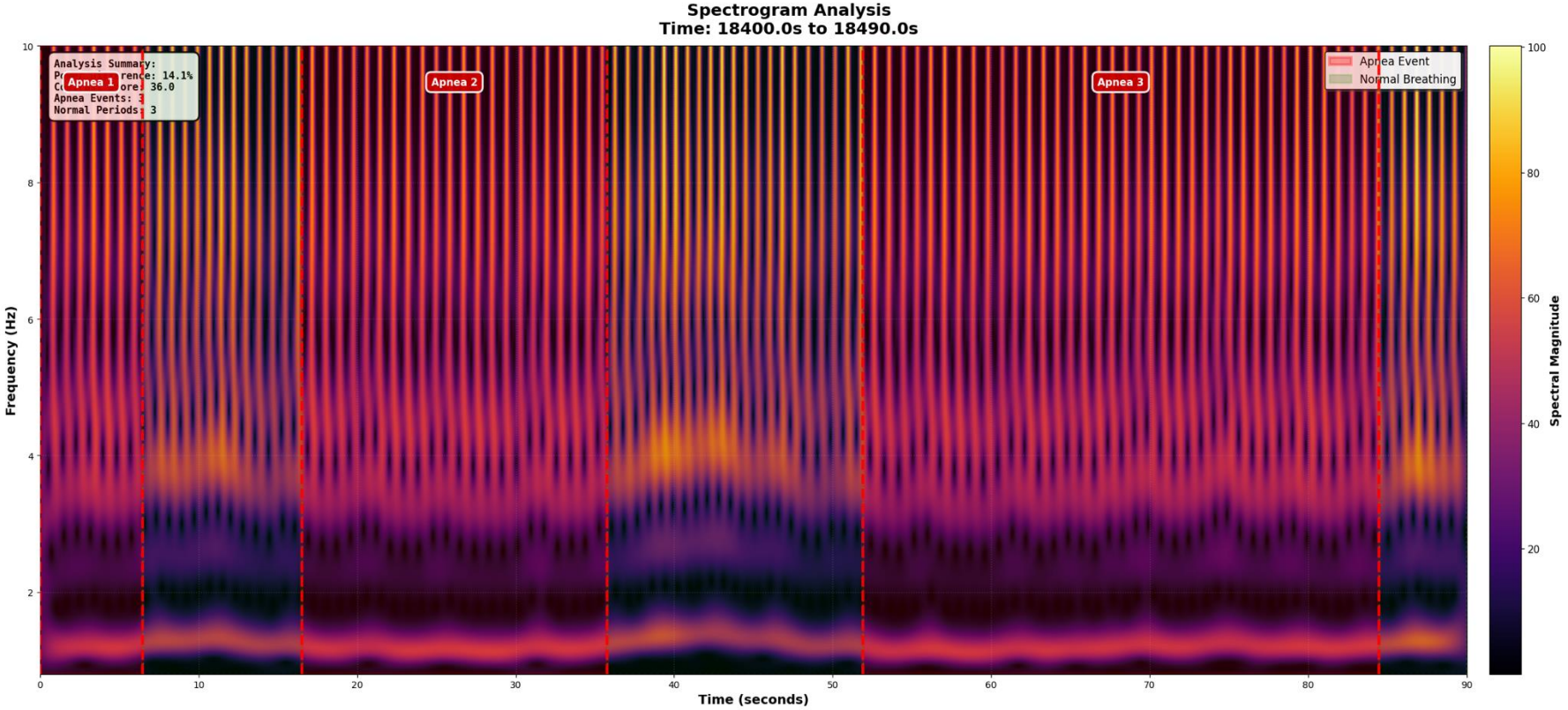


- Continuous Wavelet Transform (CWT) is a time–frequency analysis method that represents a 1D signal in both time and frequency domains.
- In this study, CWT is used to convert 1D ECG signals into 2D spectrogram representations, which serve as input for the object detection model.

ECG Spectrogram

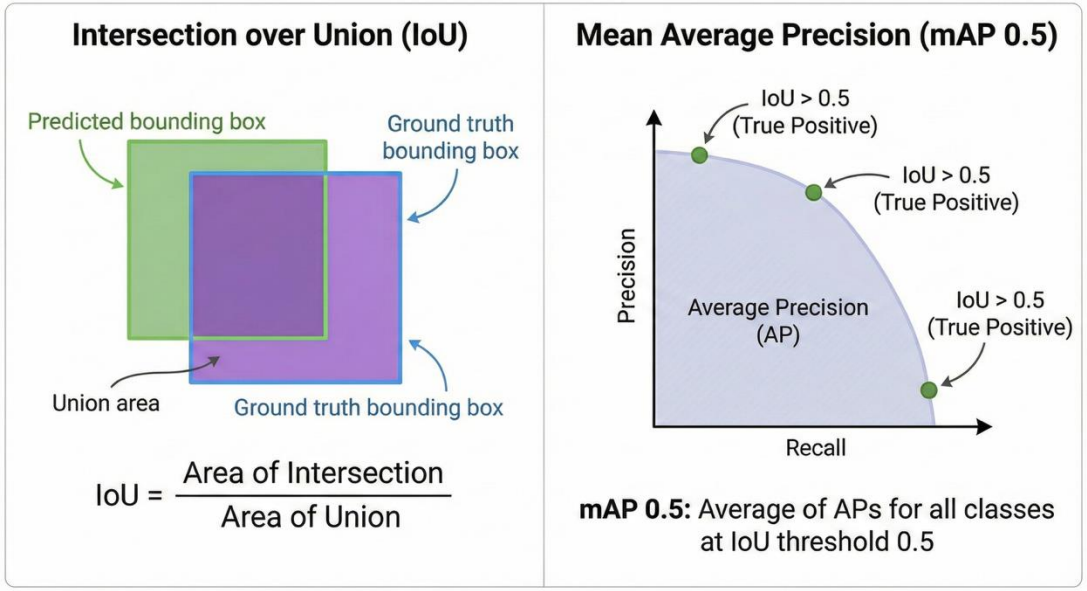


ECG Spectrogram



Evaluation Performance

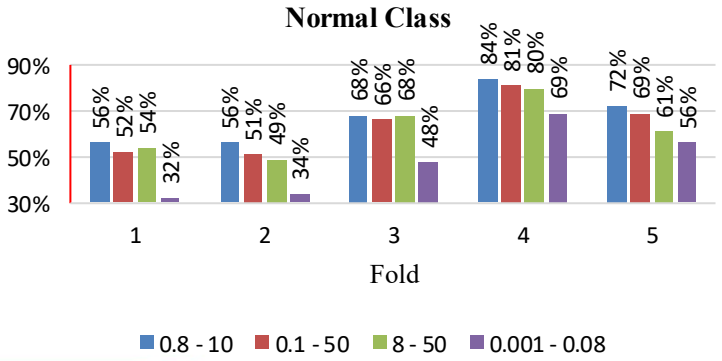
- Mean Average Precision (mAP)
  - mAP is a standard evaluation metric for object detection that measures how well a model detects and localizes objects.
  - It evaluates both correctness of detection (precision) and completeness (recall) across different confidence thresholds.
  - In this study, mAP reflects how accurately apnea events are detected as temporal objects on ECG spectrograms.

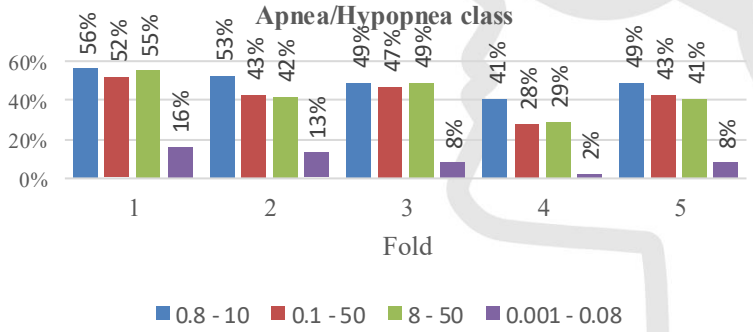


Overview subject wise 5-fold CV

Subject Wise 5-Fold CV mAP at 0.5 Comparison

Fold	Normal class				Apnea/Hypopnea class			
	0.8 - 10	0.1 - 50	8 - 50	0.001 - 0.08	0.8 - 10	0.1 - 50	8 - 50	0.001 - 0.08
1	0.56	0.52	0.54	0.32	0.56	0.52	0.55	0.16
2	0.56	0.51	0.49	0.34	0.53	0.43	0.42	0.13
3	0.68	0.66	0.68	0.48	0.49	0.47	0.49	0.08
4	0.84	0.81	0.8	0.69	0.41	0.28	0.29	0.02
5	0.72	0.69	0.61	0.56	0.49	0.43	0.41	0.08





Overview Result LOSO CV

- Compare AP score at 0.5 , Frequency range 0.1 – 50 and 0.8 - 10

Subject	Normal class		Apnea/Hypopnea class	
	0.8 - 10	0.1 - 50	0.8 - 10	0.1 - 50
1	0.7	0.66	0.35	0.22
2	0.72	0.66	0.34	0.30
3	0.46	**0.50**	0.73	**0.75**
4	0.67	0.53	0.79	0.76
5	0.59	0.50	0.33	0.22
6	0.59	**0.63**	0.88	0.85
7	0.49	0.41	0.58	**0.61**
8	0.71	0.65	0.28	0.28
9	0.6	0.57	0.6	0.60
10	0.58	0.50	0.29	**0.33**
11	0.62	**0.65**	0.28	**0.44**
12	0.31	0.25	0.58	0.47
13	0.87	0.85	0.37	**0.43**
14	0.56	0.55	0.34	0.33
15	0.54	0.45	0.49	0.43

Subject	Normal class		Apnea/Hypopnea class	
	0.8 - 10	0.1 - 50	0.8 - 10	0.1 - 50
16	0.68	0.35	0.45	**0.60**
17	0.84	**0.87**	0.22	0.39
18	0.28	0.27	0.84	0.83
19	0.86	0.58	0.56	**0.85**
20	0.58	0.52	0.54	**0.56**
21	0.57	0.56	0.61	0.56
22	0.53	**0.60**	0.24	**0.32**
23	0.75	0.59	0.53	0.31
24	0.47	0.44	0.5	0.51
25	0.58	**0.63**	0.79	**0.81**
26	1	0.99	0	0.00
27	0.94	0.86	0.31	0.23
28	0.89	0.77	0.55	0.47
29	0.6	0.58	0.65	0.64
30	0.99	0.95	0.02	0.05

Overview Result LOSO CV

- Compare mAP score at 0.5 , Frequency range 0.1 – 50 and 0.8 - 10

Subject	Normal class		Apnea/Hypopnea class	
	0.8 - 10	0.1 - 50	0.8 - 10	0.1 - 50
31	0.89	0.88	0.1	0.11
32	0.93	0.82	0.13	**0.27**
33	0.86	0.86	0.31	**0.36**
34	0.98	0.98	0.04	**0.09**
35	0.52	**0.57**	0.71	**0.72**
36	0.94	0.92	0.06	**0.09**
37	0.96	0.95	0.02	0.02
38	0.95	0.96	0.10	0.08
39	0.05	**0.38**	0.55	0.49
40	0.95	0.94	0.13	**0.21**
41	0.33	0.32	0.66	**0.71**
42	0.91	**0.92**	0.06	**0.09**
43	0.91	**0.92**	0.24	**0.44**
44	0.94	**0.95**	0.30	**0.44**
45	0.90	0.90	0.14	**0.22**

Subject	Normal class		Apnea/Hypopnea class	
	0.8 - 10	0.1 - 50	0.8 - 10	0.1 - 50
46	0.80	0.84	0.11	**0.20**
47	0.80	0.76	0.23	0.21
48	0.37	**0.48**	0.83	**0.84**
49	0.92	**0.93**	0.15	**0.27**
50	0.84	0.79	0.15	0.13

- The 0.1-50 Hz range improves apnea/hypopnea detection in 46% of subjects, with highest improvements in subjects 19 (+29%), 43 (+20%), 11 (+16%), and 44 (+14%).

Comparison mAP Performance

mAP performance of Subject-Wise 5-Fold CV

SincConv learned filter

Normal class								
Fold	**0.8 - 10**	**0.1 - 50**	**8 - 50**	**0.001 - 0.08**	**0.50 – 6.52**	**5.15 - 10.26**	**9.31 - 14.22**	**13.56 - 18.21**
1	0.56	0.52	0.54	0.32	0.50	0.53	0.54	0.54
2	0.56	0.51	0.49	0.34	0.49	0.50	0.51	0.50
3	0.68	0.66	0.68	0.48	0.63	0.63	0.63	0.66
4	0.84	0.81	0.80	0.69	0.79	0.80	0.79	0.80
5	0.72	0.69	0.61	0.56	0.69	0.71	0.69	0.69

SincConv learned filter

Apnea/Hypopnea class								
Fold	**0.8 - 10**	**0.1 - 50**	**8 - 50**	**0.001 - 0.08**	**0.50 – 6.52**	**5.15 - 10.26**	**9.31 - 14.22**	**13.56 - 18.21**
1	0.56	0.52	0.55	0.16	0.5	0.53	0.54	0.54
2	0.53	0.43	0.42	0.13	0.46	0.45	0.44	0.44
3	0.49	0.47	0.49	0.08	0.38	0.45	0.38	0.46
4	0.41	0.28	0.29	0.02	0.25	0.3	0.25	0.28
5	0.49	0.43	0.41	0.08	0.42	0.47	0.42	0.42

- SincConv-learned frequency bands show comparable but not superior mAP performance to predefined frequency ranges, with the 0.8–10 Hz band remaining the most stable across folds.

Discussion

- Under LOSO evaluation, **some subjects show improved Apnea/Hypopnea detection compared to the baseline when wider frequency ranges are used**. However, these gains are not consistent across all subjects, suggesting limited overall stability.
- In subject-wise 5-fold cross-validation, the 0.8–10 Hz frequency range provides the most stable and reliable performance for both Normal and Apnea/Hypopnea classes.
- Frequency bands learned by SincConv **achieve performance comparable to predefined ranges, but do not consistently outperform the baseline**, while very low-frequency bands perform poorly.
- Overall, adjusting the ECG frequency range may improve detection for certain individuals, but it does not consistently enhance event-based detection across subjects.

Conclusion

- Changing ECG frequency ranges **can improve detection for some subjects, but the improvements are not consistent across all patients.**
- however, the event-based detection approach remains clinically useful because it clearly shows when apnea events occur and supports event counting and AHI (Apnea Hypoapnea Index) estimation.

Limitation

- ECG signals **exhibit inherent physiological variability**, which may lead to false detections in subjects without apnea or cardiovascular abnormalities.
- Class **imbalance between normal and apnea events can bias the model** toward the normal class.
- As the focus of this study is event-based detection, performance is evaluated based on detected event counts, including detections occurring within normal-labeled segments.

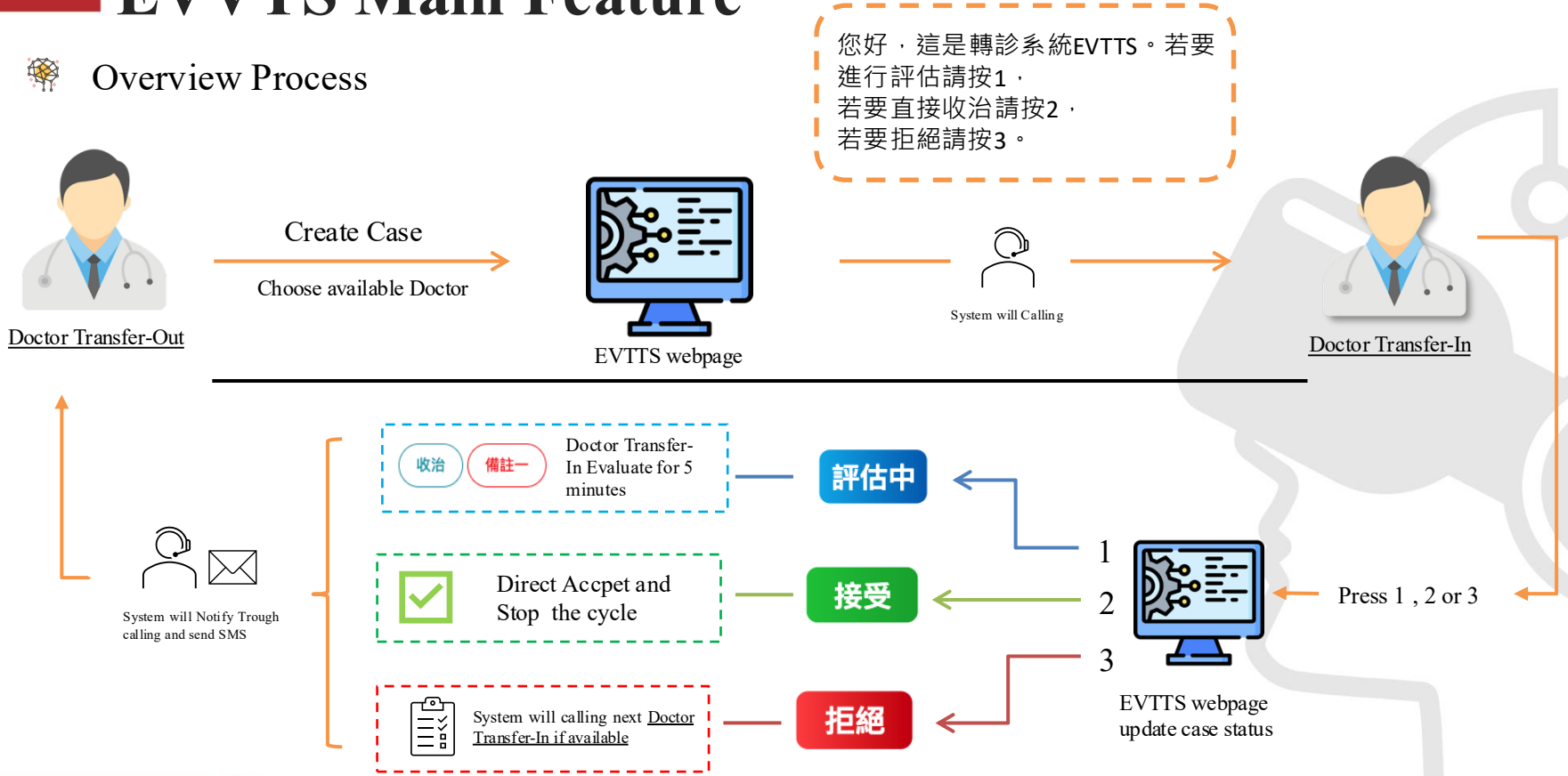
Future Work

- Explore alternative **feature generation to better capture apnea-related** patterns that are not clearly separable in current spectrogram representations, **while expanding the dataset** and applying data augmentation to improve generalizability and address class imbalance, as suggested in previous studies.

Website Work : Referral System for NCKU Hospital

EVVTS Main Feature

Overview Process



Technology Stack

- Frontend : Tailwind
- Backend: Laravel Framework
- Database: MySQL
- Communication:
  - Voice call: Chunghwa Telecom
  - SMS: TwSMS

Server & Deployment Responsibility

- Server setup and deployment
- Web service configuration
- Monitoring and maintenance

TWCC TAIWAN COMPUTING CLOUD

Thank You